# Converting Length Across Systems; Currency

*Lesson Plan*

## 1. Lesson Information

| | | | |
|---|---|---|---|
| Course | Grade 7 Accelerated (1205050) | Unit | 5 — Multi-Step Proportional Relationships |
| Lesson | Lessons 6–7 — Converting Length Across Systems; Currency | Date | _____ |
| Length | 53 minutes (one period) | Benchmark(s) | MA.7.AR.3.3 |

## 2. Standards

**MA.7.AR.3.3** Solve mathematical and real-world problems involving the conversion of units across different measurement systems.

**Must.** Convert a length between customary and metric with a printed reference-sheet ratio, either direction, in a chain; convert money with a given rate, both ways. (MA.7.AR.3.3.)

**Must not.** A ratio not on the FAST reference sheet; a real exchange rate the student must know; mass, volume or area (5.08–09).

## 3. Mathematical Thinking and Reasoning Standards

*the evidence for each, from this period*

| MTR | What it asks | The evidence in this lesson |
|---|---|---|
| MA.K12.MTR.6.1 | Assess the reasonableness of solutions. | Notes II and board 6: the estimate before the arithmetic — a centimeter is shorter than an inch, so fewer inches; a yard is nearly a meter, so about 270 cm |
| MA.K12.MTR.4.1 | Engage in discussions that reflect on the mathematical thinking of self and others. | boards 4 and 6 split; the sixty-second re-vote on which ratio |
| MA.K12.MTR.3.1 | Complete tasks with mathematical fluency. | boards 5–6: two conversions in a row, each one a form of 1, with the units cancelling on the paper |

## 4. Learning Target and Essential Question

**Learning target.** I can convert a length between the customary and metric systems with a ratio from the reference sheet, and convert money with a given rate.

**Essential question.** 1 in = 2.54 cm and 1 cm = 0.39 in are both on the sheet. How do you know which one to multiply by?

**Building on.** 5.04 — a scale read as a rate; 5.01 — multiplying by a rate; Grade 6 — converting inside one system (12 in = 1 ft).

**Working toward.** 5.08–09 — the same move on mass and volume, and squared for area; 5.10–11 — a conversion inside a rate comparison.

## 5. Sequence

*read from the deck; never typed*

| Segment | Min | What the teacher does | What students do |
|---|---|---|---|
| Settle in; post the learning target | 1 | Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet. | Copy the target. |
| Warm-Up | 6 | Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once. | Four retrieval questions, alone, no notes. Then copy the new words. |
| Notes | 10 | Teach from the front. Expand once per law; say the OFF line out loud. | Copy the notes; answer the checks aloud. |
| Example 1 | 4 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Example 2 | 4 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Whiteboards | 11 | One question at a time. Boards down until the cue, all up together; scan the back row first. | Answer on a board; hold it up on three. |
| Independent Set | 6 | Silent. Circulate and mark what you see; do not teach. | Six questions, written, alone. |
| IXL | 5 | Assign the day's IXL skills. | Start the skills; finish tonight. |
| **Total** | **53** | | |

## 6. Gradual Release

**I do.** The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

**We do.** Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

**You do.** The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

## 7. Higher-Order Questions

| Question | DOK |
|---|---|
| Why does multiplying 15 in by 2.54 cm per inch change the unit and not the length? | DOK 2 |
| Jo turned 30 cm into 76.2 in. Without computing, how do you know that is wrong? | DOK 3 |
| Money converts like length, but the ratio is on the problem and not on the sheet. Why? | DOK 2 |

## 8. Checks for Understanding

*each board, and what to do about it*

| # | What the board asks | Answer | If the board is wrong |
|---|---|---|---|
| 1 | $1\text{ in} = 2.54\text{ cm}$ . A ruler is 20 in long. How many centimeters is that? | ***50.8 cm*** | 7.87 — divided by 2.54 [Notes II: centimeters are smaller, so more of them]; 22.54 — added the ratio |
| 2 | $1\text{ cm} = 0.39\text{ in}$ . A phone is 20 cm long. How many inches is that? | ***7.8 in*** | 50.8 — multiplied by 2.54 [Notes II: pick the ratio whose 1 is the unit you have]; 0.78 — place value |
| 3 | $1\text{ mi} = 1.61\text{ km}$ . A race is 5 mi long. How many kilometers is that? | ***8.05 km*** | 3.1 — multiplied by 0.62, the other ratio [Notes II]; 6.61 — added 1.61 |
| 4 | $1\text{ km} = 0.62\text{ mi}$ . A trail is 5 km long. How many miles is that? | ***D — 3.1 mi*** | (A) 5 + 0.62: the additive error [7.AR.3.3: Math Nation's named misconception]; (B) place value: 5 × 0.62 is 3.1, not 0.31 [4.03]; (C) × 1.61, the ratio for the other direction [B1G-M 7.AR.3.3: the inverted ratio; Notes II] |
| 5 | $1\text{ ft} = 12\text{ in}$ and $1\text{ in} = 2.54\text{ cm}$ . A door is 6 ft tall. How many centimeters is that? | ***182.88 cm*** | 15.24 — 6 × 2.54, the feet treated as inches [Notes I: the unit you have must match the ratio's 1]; 183 — rounded early (accept with the work shown) |
| 6 | $1\text{ yd} = 3\text{ ft}$ , $1\text{ ft} = 12\text{ in}$ , $1\text{ in} = 2.54\text{ cm}$ . A table is 3 yd long. About how many centimeters is that? Estimate, then compute. | ***About 270; exactly 274.32 cm*** | 7.62 — 3 × 2.54, the yards treated as inches [Notes II: estimate first — a 3-yd table is not 8 cm]; 91.44 — stopped at 36 in × 2.54 (one yard) |
| 7 | Rate on Nov 2: $\$1 = 1.32\text{ CAD}$ ; $1\text{ CAD} = 16.2\text{ pesos}$ . Theo changes $200 to CAD, then to pesos. How many pesos does he get? | ***4,276.80 pesos*** | 3,240 — 200 × 16.2, the first hop skipped [Notes III: the rate's 1 must be the currency you have]; 2,454.55 — divided by 1.32 on the first hop |
| 8 | $1\text{ km} = 0.62\text{ mi}$ . Theo, who weighs 110 lb, rides a 12 km trail. How long is the trail in miles? | ***7.44 mi*** | 19.32 — × 1.61 [B1G-M 7.AR.3.3: the inverted ratio]; 49.94 kg — converted the weight instead |
| 9 | $1\text{ in} = 2.54\text{ cm}$ and $1\text{ cm} = 0.39\text{ in}$ . Jo converts 30 cm to inches by multiplying by 2.54 and gets 76.2 in. Name the error and fix it. | ***She used the inches ratio on centimeters. Inches are bigger, so fewer: 11.7 in.*** | 'She should have used 0.39' with no reason — the fix without the error named; 76.2 defended as 'more inches' — the estimate backwards [B1G-M 7.AR.3.3] |

**MTR.4.1 — the one peer move.** When a board splits — no option holding about two-thirds of the room — sixty seconds of "convince the person next to you", then re-vote, then reveal. The minute comes out of the whiteboard remainder.

## 9. Differentiation

**ESE / 504.** The reference sheet's two conversion lines stay on the board all period. Every board starts by pointing at the ratio whose 1 is the unit you have, then multiplying. The estimate is said aloud before the calculator.

**ELL.** **Customary** is the system with inches and miles; **metric** has centimeters and kilometers. The word **per** in "2.54 cm per inch" is the same per as miles per hour.

**Enrichment.** Board 7 finishes early for some: ask them to run the chain backwards — pesos to Canadian dollars to US dollars — and say which operation each step became.

## 10. Closure, Vocabulary, Materials, Practice

**Closure.** Board 9 is written work and it is the exit evidence: the error named (Jo multiplied by the ratio whose 1 is inches, when she had centimeters), the estimate that catches it (inches are longer, so fewer of them), and the fix (30 × 0.39 = 11.7 in). A board that writes the right number with no sentence about the direction has not met the lesson.

**Vocabulary.** **customary**; **metric**; **conversion ratio**; **exchange rate**

**Materials.** Whiteboards and markers. The FAST reference sheet's conversion lines on the board all period (they are printed on every item as well). Calculators are allowed on everything and printed on nothing.

**Practice.** The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Convert between customary and metric systems — E8Z; Compare and convert customary units — 4B8; Compare and convert metric units — HW2.